

## Vehicle Stability Metrics

### Stability Index:

- The static directional stability of the vehicle.
- If negative it means that a lateral force/acceleration disturbance applied to the car at the CG will result in a yaw moment that turns the car away from the disturbance. This is stable behaviour.
- If the value is positive, the vehicle is unstable, it will turn itself towards the disturbance, further increasing the lateral acceleration and yaw moment.
- It is generally possible (but difficult) to drive a car that has a positive SI without spinning, because of the finite yaw response time of a vehicle with finite yaw inertia. The driver can induce a yaw control moments using steering, brake and throttle to continually correct the destabilising moment, keeping the yaw acceleration of the vehicle under control.
- This number is the slope of the Milliken Moment Method Diagram constant-steer line, its value is normalised yaw moment per G of lateral acceleration (1/g).
- The yaw moment is normalised by wheelbase and vehicle weight, meaning it is nominally comparable between vehicles of different types.
- The value is comprised of the understeer gradient derivative (straightline stability derivative), and the yaw-damping (path curvature stiffness) derivative.
- The path curvature stiffness derivative is the stabilising yaw moment that results from the fact that the car is traversing a curved path, which induced slip angles that try to straighten the path of the vehicle.

### Spin Tendency:

- This is the normalised yaw moment when both front and rear tyres are at their slip peaks.
- A positive number means the vehicle will tend to instability when lateral acceleration approaches the maximum trimmed lateral acceleration.
- Spin Tendency is important, because it indicates what the vehicle stability tends to do if the vehicle is pushed to the limit, whereas the stability index tells you what the stability actually is for the current vehicle state, which may be far from the limit.
- When spin tendency is positive, the car is rear-grip limited.
- The value is normalised, making it nominally comparable between different types of cars.

### Available Control:

- This is the additional available yaw moment from the front wheels, if the driver were to step-steer to the peak front slip angle.
- It is a measure of how much control the driver has to tighten the path radius.
- The number is normally positive, but never negative. When the number is zero, the front axle is saturated.

### Available Steer:

- This is the additional road-wheel angle that can be added to get to the peak of the front tyres.
- This number is normally positive when the fronts are slipping less than the peak.
- When the driver steers beyond the peak of the front tyres, the number will be negative.
- This number is interesting to compare with Available Control, because even though both should become zero at the same time, it is normal that available control will be very small when there is still quite a lot of available steer left. In this situation it means the car has a relatively flat-topped

front slip curve. When drivers are fast, but they do not wear their front tyres very much, they are normally exploiting this “flat-topped” characteristic.

#### **Maximum Trimmed Lateral Acceleration:**

- This is the lateral acceleration at which one of the ends of the car will reach its peak.
- When spin tendency is positive, the rear tyres will dictate this acceleration. When spin tendency is negative, the front tyre peak will be reached first.
- Generally this maximum acceleration cannot be reached when spin tendency is positive, because the car becomes unstable (SI positive), before the limit of the rear tyre is reached.
- The maximum can be slightly exceeded for short periods of time in reality, because this is a steady-state trimmed value.
- This value is arguably the key metric that can be used to determine the limit (mid) cornering speed of the vehicle, assuming that the vehicle has negative spin tendency.
- AVL recommends overlaying this value with the actual lateral acceleration (in the track frame).

#### **Maximum Stable Lateral Acceleration:**

- This provides a “better” performance limit measure than the Maximum Trimmed Lat Acc, because it reports a maximum “driveable” lateral acceleration when the vehicle has a positive spin tendency.
- When the car has a negative spin tendency, it reports the same value as the Maximum Trimmed Lateral Acceleration, because the vehicle will be by definition stable at the limit.

